Milestone 4 Technical Summary

Architecture and End-to-End Behavior

The prototype supports a graduate business career advisor who screens job opportunities for MSBA international students. It accepts a free-text query or posting, retrieves comparable real postings from an 80,000-row training index, applies a transparent four-label role-fit rubric, and returns a recommendation with evidence. This design uses retrieval-augmented generation as a grounding pattern: generation is a constrained template, and every job-specific claim comes from retrieved fields.

Layer	Method	Why it matters
Input	Advisor query or raw job text	Matches the real screening workflow
Retrieval	Sparse index, top-6 over 80,000 postings	Supplies comparable source evidence
Decision	Transparent role-fit rubric	Keeps the triage logic inspectable
Generation	Evidence-bound response template	Prevents unsupported job facts
Review	Advisor confirms before forwarding	Limits automation harm

Preliminary Evaluation vs. Baseline

System	Set	Accuracy	Macro F1
M2 TF-IDF baseline	20k held-out	79.3%	78.5%
Hashed centroid baseline	same 4k	79.7%	79.1%
M4 grounded prototype	same 4k	80.0%	80.1%

The same-set comparison shows a preliminary macro-F1 gain of +0.009. Retrieval succeeds for 100.0% of evaluation inputs, and support hit@5 is 90.0%. The full M2 result is context only because it uses a larger validation set.

What the Errors Mean

The prototype makes 798 errors on 4,000 balanced validation rows. High-fit F1 is 0.960, but medium-fit F1 is 0.726. The two dominant confusions are medium_fit -> low_fit (380 rows) and unclear -> low_fit (254 rows). A production pilot should therefore optimize high-fit precision and route medium/unclear cases to review instead of treating low-fit as an automatic rejection.

Grounded Case Evidence

Workflow case	Query label	Retrieved evidence pattern	Operational response
entry level analytics	high_fit	high_fit: 5	Forward candidates after advisor review
senior engineering filter	low_fit	low_fit: 2, medium_fit: 2, unclear: 1	Hold a role that is too senior or engineering-heavy
authorization missing	unclear	high_fit: 4, medium_fit: 1	Escalate missing CPT/OPT or sponsorship evidence
thin metadata	unclear	high_fit: 1, medium_fit: 4	Investigate an incomplete posting

Grounding and Hallucination Analysis

All 4 cases returned source evidence, covering 20 inspected records. One retrieved record contained explicit negative authorization language, which the system surfaced. The generated outputs contain zero unsupported authorization claims and attach an authorization verification caveat in 100% of cases.

Risk 1: authorization misinformation

The public dataset does not reliably contain CPT, OPT, H-1B, or sponsorship language. A wrong claim could cause a student to spend time on an ineligible role or miss a viable opportunity. Mitigation: do not infer eligibility, show the missing-evidence caveat, and require advisor review.

Risk 2: weak-label bias and automation harm

Labels are derived from transparent rules rather than final advisor judgments. They can over-penalize engineering vocabulary and incomplete postings. Mitigation: expose retrieved evidence, retain the human decision, report per-label errors, and collect de-identified advisor corrections for the next evaluation.

Recommendation

Use the prototype for first-pass prioritization, not automatic acceptance or rejection. Pilot it on de-identified Career Center postings, measure high-fit precision and advisor time saved, and add a separate authorization extractor only after reliable source text is available.